UNIVERSITI TEKNOLOGI MALAYSIA

# ASSIGNMENT <Test1 & Test2>

## Topic: <Chapter 1,2,3,4>

## COMPUTATIONAL MATHEMATICS

## Sem 1 Session 2021/2022

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|---|---|
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| DATE SUBMITTED: | <26 May 2022>, if late, put reason |
| MARKS | <leave blank> |

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Question 1

Given the following linear system:

$$X - Y - 2Z = 3$$
$$-X + 2Y + 3Z = 1$$
$$2X - 2Y - 2Z = -2$$

a) form A coefficient matrix A for the linear system

$$\begin{matrix} A & & X & = & B \end{matrix}$$
$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

b) find inverse of A, $(A^{-1})$ using cofactor matrix

use: row 1 expansion to obtain the determinant

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \\ 2 & -2 & 2 \end{bmatrix} \quad \begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

$= 2 - 4 + 4$

$|A|$ $\det A = 2$ ✱

$$= 1\begin{bmatrix} 2 & 3 \\ -2 & -2 \end{bmatrix} - (-1)\begin{bmatrix} -1 & 3 \\ 2 & -2 \end{bmatrix} + (-2)\begin{bmatrix} -1 & 2 \\ 2 & -2 \end{bmatrix}$$

$$= 1((2)(-2) - (3)(-2)) + 1((-1)(-2) - (3)(2)) - 2((-1)(-2) - (2)(2))$$

$$= 1(-4+6) + 1(2-6) - 2(2-4)$$

$$= 1(2) + 1(-4) - 2(-2)$$

b) Cofactor:

$$\begin{bmatrix} +\begin{bmatrix} 2 & 3 \\ -2 & -2 \end{bmatrix} - \begin{bmatrix} -1 & 3 \\ 2 & -2 \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ 2 & -2 \end{bmatrix} \\ -\begin{bmatrix} -1 & -2 \\ -2 & -2 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix} \\ +\begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} ((2)(-2) - (3)(-2)) & -((-1)(-2) - (3)(2)) & +((-1)(-2) - (2)(2)) \\ -((-1)(-2) - (-2)(-2)) & +((1)(-2) - (-2)(2)) & -((1)(-2) - (-1)(2)) \\ +((-1)(3) - (-2)(2)) & -((1)(3) - (-2)(-1)) & +((1)(2) - (-1)(-1)) \end{bmatrix}$$

$$= \begin{bmatrix} -4+6 & -2-6 & 2-4 \\ -2-4 & -2+4 & 2+2 \\ -3+4 & -3-2 & 2-1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -8 & -2 \\ -6 & 2 & 4 \\ 1 & -5 & 1 \end{bmatrix}$$

b)

$$\frac{1}{|A|}\begin{bmatrix} 2 & -6 & 1 \\ -8 & 2 & -5 \\ -2 & 4 & 1 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 2 & -6 & 1 \\ -8 & 2 & -5 \\ -2 & 4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{2} & \frac{-6}{2} & \frac{1}{2} \\ \frac{-8}{2} & \frac{2}{2} & \frac{-5}{2} \\ \frac{-2}{2} & \frac{4}{2} & \frac{1}{2} \end{bmatrix}$$

inverse:

$$= \begin{bmatrix} 1 & -3 & \frac{1}{2} \\ -4 & 1 & \frac{-5}{2} \\ -1 & 2 & \frac{1}{2} \end{bmatrix}$$

c) the general form of LU matrix is given as follow. Find the solution of the following linear equation using LU factorization in Crout form matrix

$$x - y - 2z = 3$$
$$-x + 2y + 3z = 1$$
$$2x - 2y - 2z = -2$$

$$\underset{A}{\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \\ 2 & -2 & -2 \end{bmatrix}} \underset{B}{\begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}}$$

$$\underset{A}{\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \\ 2 & -2 & -2 \end{bmatrix}} = \underset{L}{\begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix}} \underset{U}{\begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}}$$

$$= \begin{bmatrix} L_{11} & L_{11}U_{12} & L_{11}U_{13} \\ L_{21} & L_{21}U_{12} + L_{22} & L_{21}U_{13} + L_{22}U_{23} \\ L_{31} & L_{31}U_{12} + L_{32} & L_{31}U_{13} + L_{32}U_{23} + L_{33} \end{bmatrix}$$

$L_{11} = 1$ $\quad L_{11}U_{12} = -1$ $\quad L_{11}U_{13} = -2$

$1\,U_{12} = -1$ $\quad 1\,U_{13} = -2$

$U_{12} = -1 - 1$ $\quad = \frac{-2}{1}$

$= -2$ $\quad = -2$

c) $L_{21} = -1$ $\quad L_{21}U_{12} + L_{22} = 2$ $\quad L_{21}U_{13} + L_{22}U_{23} = 3$

$-1(-2) + L_{22} = 2$ $\quad -1(-2) + 0\,U_{23} = 3$

$2 + L_{22} = 2$ $\quad 2 + 0\,U_{23} = 3$

$L_{22} = 2 - 2$ $\quad U_{23} = 3 - 2 - 0$

$= 0$ $\quad = 1$

$L_{31} = 2$ $\quad L_{31}U_{12} + L_{32} = -2$ $\quad L_{31}U_{13} + L_{32}U_{23} + L_{33} = -2$

$2(-2) + L_{32} = -2$ $\quad 2(-2) + 2(1) + L_{33} = -2$

$-4 + L_{32} = -2$ $\quad -4 + 2 + L_{33} = -2$

$L_{32} = -2 + 4$ $\quad -2 + L_{33} = -2$

$= 2$ $\quad = -2 + 2$

$= 0$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 2 & 2 & 0 \end{bmatrix} \qquad U = \begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L \quad Y = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

$Y_1 = 3$

$-Y_2 = 1$

$2Y_1 - 2Y_2 = -2$

$-1\,Y_2 = 1$

$Y_2 = \frac{1}{-1}$

$= -1$

$2Y_1 - 2Y_2 = -2$

$2(3) - 2(-1) = -2$

$6 + 2 = -2$

$8 = -2$

Intermediate solution:

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$$

$$U \quad X = Y$$

$$\begin{bmatrix} 1 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$$

$X_3 = -2$

$X_2 - X_3 = -1$

$X_2 - (-2) = -1$

$X_2 + 2 = -1$

$X_2 = -1 - 2$

$= -3$

$X_1 - (-2X_2) + (-2X_3) = 3$

$X_1 + 2X_2 - 2X_3 = 3$

$X_1 + 2(-3) - 2(-2) = 3$

$X_1 - 6 + 4 = 3$

$X_1 = 3 + 6 - 4$

$= 5$

System solution:

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ -2 \end{bmatrix}$$

Question 2

a) Given $u = (-4, 0, 2, -2)$, $v = (2, 0, -1, 1)$

i. Find $\|2u\| - 2\|u\|$ $= \sqrt{(2\cdot-4)^2+(2\cdot0)^2+(2\cdot2)^2+(2\cdot-2)^2} - 2\sqrt{(-4^2+0^2+2^2+(-2)^2)}$

$= \sqrt{64+0+16+16} \quad - 2\sqrt{(+16+0+4+(+4))}$

$= \sqrt{96} \quad - 2\sqrt{24}$

$= 4\sqrt{6} - 2\sqrt{6}$

$= 0$

ii. Find the dot product of vectors u and v

$(-4, 0, 2, -2) \cdot (2, 0, -1, 1)$

$= ((-4)(2)) + ((0)(0)) + ((2)(-1)) + ((-2)(1))$

$= -8 + 0 + (-2) + (-2)$

$= -12$

iii. Find $5(v + 3u) - 3v$

$= 5((2, 0, -1, 1) + 3(-4, 0, 2, -2)) - 3(2, 0, -1, 1)$

$= 5((2, 0, -1, 1) + (-12, 0, 6, -6)) - 3(2, 0, -1, 1)$

$= 5(-10, 0, 5, -5) - 3(2, 0, -1, 1)$

$= (-50, 0, 25, -25) - (6, 0, -3, 3)$

$= (-56, 0, 28, -28)$

b. Find the distance between vector $u = (0,2,2)$ and $v = (2,0,1)$

$$d(\bar{u},\bar{u}) = \sqrt{(0-2)^2 + (2-0)^2 + (2-1)^2}$$

$$= \sqrt{-2^2 + 2^2 + 1^2}$$

$$= \sqrt{4 + 4 + 1}$$

$$= \sqrt{9}$$

$$= 3$$

C Given $u = (5,-1,-2)$ and $v = (-2,4,-2)$, complete $u \times v$

$$\begin{array}{cccccc} 5 & -1 & -2 & 5 & -1 & -2 \\ -2 & 4 & -2 & -2 & 4 & -2 \end{array}$$

$$2-(-8) \quad 4-(-10) \quad 20-2$$

$$u \times v = (10, \quad 14, \quad 18)$$

Question 3

Let $M_1 = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$, $M_2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $M_3 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

a) Show that the matrices are linearly independent.

$$= C_1\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} + C_2\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + C_3\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= C_1 + C_2 = 0$$

$$C_2 + C_3 = 0$$

$$-C_1 + C_2 + C_3 = 0$$

$$C_3 = 0$$

$C_3 = 0$ :-

From (2), $C_2 = 0$

From (3), $C_1 = 0$

Therefore the matrixes $M_1, M_2, M_3$ are linearly independent.

$$= \left[\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$R_3 \times R_4 \longrightarrow R_3$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$R_4 \times R_1 \longrightarrow R_4$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$= C_1 + C_2 = 0 \quad -(1)$$

$$C_2 + C_3 = 0 \quad -(2)$$

$$C_3 = 0 \quad -(3)$$

b) Find the unique scalars $c_1, c_2, c_3$ such that the matrix $M = \begin{bmatrix} 3 & 5 \\ 4 & 3 \end{bmatrix}$ can be written as $M = c_1M_1 + c_2M_2 + c_3M_3$

$$c_1M_1 + c_2M_2 + c_3M_3 = M$$

$$c_1\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} + c_2\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + c_3\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} c_1 + c_2 + 0 = 3 & 0 + c_2 + c_3 = 5 \\ -c_1 + c_2 + c_3 = 4 & 0 + 0 + c_3 = 3 \end{bmatrix}$$

$$c_1 + c_2 = 3 \qquad c_2 + c_3 = 5$$

$$-c_1 + c_2 + c_3 = 4 \qquad c_3 = 3$$

c) Show that the matrix $M = \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$ cannot be written as a linear combination of $M_1, M_2, M_3$

$$c_1M_1 + c_2M_2 + c_3M_3 = \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$$

Show that there is no solution for $c_1, c_2, c_3$

d) Given $M_4 = \begin{bmatrix} 0 & 1 \\ 1 & d \end{bmatrix}$ what is the value of $d$ so that $M_2, M_3, M_4$ are linearly independent.

$$c_1M_2 + c_2M_3 + c_3M_4 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$c_1\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + c_2\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + c_4\begin{bmatrix} 0 & 1 \\ 1 & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} c_1 & c_1 + c_2 + c_4 \\ c_1 + c_2 + c_4 & c_2 + dc_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$c_1 = 0$

$c_2 + c_4 = 0$

$c_2 + dc_4 = 0$

$d = 1$ ✱

i) Let, W be the subset of vector space of 2×2 matrices

$$W = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \middle| \; a+b=5 \right\}$$

Check whether W is subspace under the inherited operations

W is non empty.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin W$$

W is not a subspace

Test 2

Question 1

i) Let, W be the subset of vector space of 2×2 matrices

$$W = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \middle| \; a+b=5 \right\}$$

Check whether W is subspace under the inherited operations

using this theorem:

$\overline{0} \in W$

for $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in W$, must have $a = 0$

$b = 0$

but $a+b=5$

$\therefore \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin W$

$\therefore$ W is not a subspace

ii) Let, $B_1 = \{V_1, V_2\}$ and $B_2 = \{W_1, W_2\}$ be bases for a vector space,

$V = R^2$, and suppose

$V_1 = 2W_1 + 3W_2$

$V_2 = -W_2$

a) State the transition matrix from $B_1$ to $B_2$, $\left[I\right]_{B_2}^{B_1}$

$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

iii) Matrix A is given as follows

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \end{bmatrix}$$

$\bar{v}_1 \quad \bar{v}_2 \quad \bar{v}_3 \quad \bar{v}_4$

a) State the row space of A

Column (A) = span ( $\{\bar{v}_1, \bar{v}_2, \bar{v}_3, \bar{v}_4, \ldots, \bar{v}_n\}$ )

$$\text{Column}(A) = \text{span}\left(\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\2\\1\end{bmatrix}\right)$$

$R_3 - 1(R_1) \longrightarrow R_3$

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$R_3(R_1) \longrightarrow R_3$

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Basis for Column (A) $= \left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\2\end{bmatrix}\right\}$

$\bar{b}_1 \quad \bar{b}_2 \quad \bar{b}_3$

Column (A) is 3 dimensional

Column (A) $< R^3$

iii) Matrix A given as follows

$$A=\begin{bmatrix}1&0&1&1\\0&1&1&2\\1&1&2&1\end{bmatrix}$$

c) Find the basis of column space of A

c) $A \sim B \Rightarrow$ select non-zero rows of B

$R_3(R_2) \longrightarrow R_3$

$$\begin{bmatrix}1&0&1&1\\0&1&1&2\\0&1&2&2\end{bmatrix}$$

$R_3 - R_2 \longrightarrow R_3$

$$\begin{bmatrix}1&0&1&1\\0&1&1&2\\0&0&1&0\end{bmatrix}$$

$R_3 - R_2(0) \longrightarrow R_3$

$$\begin{bmatrix}1&0&1&1\\0&1&1&2\\0&0&0&0\end{bmatrix}$$

Basis for Row(A)

$$\left\{\begin{bmatrix}1\\0\\1\\1\end{bmatrix},\begin{bmatrix}0\\1\\1\\2\end{bmatrix}\right\}$$

b) find the nullspace of A

Nullspace of A = $N(A) \longrightarrow [A \mid 0]$

$$\underset{A}{\begin{bmatrix}1&0&1&1\\0&1&1&2\\1&1&2&1\end{bmatrix}}\underset{X}{\begin{bmatrix}X_1\\X_2\\X_3\\X_4\end{bmatrix}}=\underset{0}{\begin{bmatrix}0\\0\\0\end{bmatrix}}$$

$R_3(R_2) \longrightarrow R_3$

$$\left[\begin{array}{cccc|c}1&0&1&1&0\\0&1&1&2&0\\0&1&2&2&0\end{array}\right]$$

$R_3 - R_2 \longrightarrow R_3$

$$\left[\begin{array}{cccc|c}1&0&1&1&0\\0&1&1&2&0\\0&0&1&0&0\end{array}\right]$$

$R_1 - R_3 \longrightarrow R_1$

$$\left[\begin{array}{cccc|c}1&0&0&1&0\\0&1&1&2&0\\0&0&1&0&0\end{array}\right]$$

$R_3(R_1) \longrightarrow R_3$

$$\left[\begin{array}{cccc|c}1&0&0&1&0\\0&1&1&2&0\\0&0&0&0&0\end{array}\right]$$

$$X_1 - X_4 = 0$$

$$-X_2 + X_3 - 2X_4 = 0$$

$$X_1 = X_4$$

$$-X_2 = -X_3 - 2X_4$$

$$X_3 = X_3$$

$$X_4 = X_4$$

$$N(A) = \begin{bmatrix}X\\-X-2X\\X\\X\end{bmatrix} \quad \text{Basis} = \begin{bmatrix}1\\-1-2\\1\\1\end{bmatrix}$$

d) Find the rank and nullity of A

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 \end{bmatrix}$$

Reduced row

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

pivot columns = 2

Non-pivot columns = 2

Rank A : 2

Nul A : 2

Question 2

i) Find function $T: R^3 \rightarrow R^2$ is defined as

$$T(v) = A(v) = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$2 \times 3$

$3 \times 1$

a) Find the image of $v = \begin{bmatrix} -5 \\ 7 \\ -3 \end{bmatrix}$

$$T\begin{bmatrix} -5 \\ 7 \\ -3 \end{bmatrix} = \begin{bmatrix} 1v_1 + (-1)v_2 + (-2)v_3 \\ -1v_1 + 2v_2 + 3v_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-5) + (-1)(7) + (-2)(-3) \\ -1(-5) + 2(7) + 3(-3) \end{bmatrix}$$

$$= \begin{bmatrix} -6 \\ 10 \end{bmatrix}$$

b) Find the kernel of the linear transformation, $\ker(T)$

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$v_1 - t = 0$

$v_2 - t = 0$

$\ker(T) = \{(t, t, t),\ t \in R\}$

basis $= \{(1, 1, 1)\}$

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ -1 & 2 & 3 & 0 \end{array}\right]$$

$R_2 + R_1 \rightarrow R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$R_1 + R_3 \rightarrow R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

let $v_3 = t$

Question 3

Given the function $T: R^2 \to R^3$ defined as

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ 2x_2 - x_1 \\ 4x_1 \end{bmatrix}$$

i) Show that T is a linear transformation

$$T(\bar{u} + \bar{v}) = T(\bar{u}) + T(\bar{v})$$

$$\bar{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad \bar{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right) = \left(\begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}\right)$$

$$\begin{bmatrix} a_1 + b_1 - (a_2 + b_2) \\ a_1 + b_1 + (2b_2 - a_2) \\ 4(a_1 + b_1) \end{bmatrix}$$

$$\begin{bmatrix} a_1 + b_1 - a_2 - b_2 \\ a_1 + a_2 + b_1 + b_2 \\ 4a_1 + 4b_1 \end{bmatrix} = T(\bar{u}) + (\bar{v})$$

$$T(\bar{u}) + T(\bar{v})$$

$$T\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) + T\left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}\right)$$

$$\begin{bmatrix} a_1 - a_2 \\ a_1 + a_2 \\ 4a_1 \end{bmatrix} + \begin{bmatrix} b_1 - b_2 \\ b_1 + b_2 \\ 4b_1 \end{bmatrix}$$

$$\begin{bmatrix} a_1 - a_2 + b_1 - b_2 \\ a_1 + a_2 + b_1 + b_2 \\ 4a_1 + 4b_1 \end{bmatrix} = T(\bar{u} + \bar{v})$$

$T(\bar{u} + \bar{v}) = T(\bar{u}) + T(\bar{v})$ T is a linear transformation.